

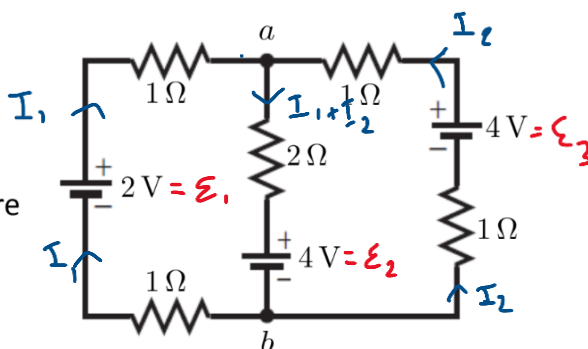
Latihan Soal UAS Fisika

1. Basic Resistor Multiloop Circuit

Latihan

1. Tentukanlah

- a) Besar dan arah arus listrik pada masing-masing batere
- b) Beda potensial $V_b - V_a$
- c) Daya pada hambatan $2\ \Omega$



Jawab:

Loop Kiri

$$\begin{aligned}
 2 + I_1(1 + 1 + 2) + I_2(2) &= 0 \\
 4I_1 + 2I_2 &= -2 \\
 I_2 &= -1 - 2I_1 \dots\dots (1)
 \end{aligned}$$

Loop Kanan

$$\begin{aligned}
 4 - 4 + I_2(1 + 1 + 2) + I_1(2) &= 0 \\
 4I_2 + 2I_1 &= 0 \\
 I_1 &= -2I_2 \dots\dots (2)
 \end{aligned}$$

Substitusi (2) ke (1)

$$\begin{aligned}
 I_2 &= -1 - 2(-2I_2) \\
 I_2 &= -1 + 4I_2 \\
 1 &= 3I_2 \\
 I_2 &= \frac{1}{3} \text{ A}
 \end{aligned}$$

Substitusi I_2 ke (2)

$$I_1 = -2\left(\frac{1}{3}\right) = -\frac{2}{3} \text{ A}$$

Besar dan Arah Arus Listrik Pada Masing-Masing Baterai

Baterai 1

$$I = \frac{2}{3} A \text{ (dari kutub positif ke kutub negatif)}$$

Baterai 2

$$I = \frac{1}{3} A \text{ (dari kutub negatif ke kutub positif)}$$

Baterai 3

$$I = \frac{1}{3} A \text{ (dari kutub negatif ke kutub positif)}$$

Beda Potensial $V_b - V_a$

$$V_a - \varepsilon_2 + IR = V_b$$

$$-4 + \left(-\frac{1}{3}\right) = V_b - V_a$$

$$-\frac{13}{3} V = V_b - V_a$$

Daya Pada Hambatan 2Ω

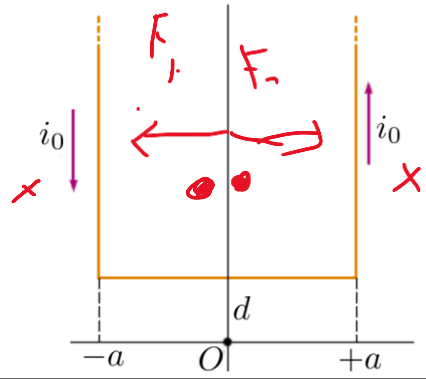
$$P = I^2 R$$

$$P = \left(-\frac{1}{3}\right)^2 (2)$$

$$P = \frac{2}{9} \text{ Watt}$$

2. Magnetic Field and Force Cases

Suatu kawat yang panjang dibentuk menjadi seperti gambar. Tentukan besar dan arah medan magnet di titik O jika kawat dialiri arus sebesar i_0



Jawab:

$$B = \frac{\mu_0 I}{2\pi a}$$

$$B = \frac{4\pi \times 10^{-7} I_0}{2\pi a}$$

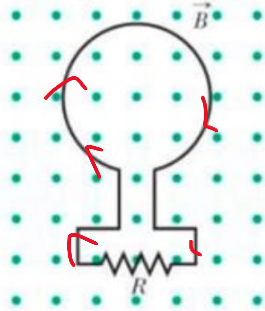
$$B = \frac{10^{-7} I_0}{a}$$

$$B_{total} = 2B$$

$$B_{total} = \frac{2 \times 10^{-7} I_0}{a} T$$

3. Induced Current by Inductor

Gunakan gambar dibawah untuk menjawab pertanyaan berikut



Nilai *magnetic flux* pada *loop* kabel berdasarkan persamaan $\Phi_B = 13t^2 + 4t + 8$ milliWeber, dan t dalam satuan *second*.

- Tentukan nilai emf induksi pada saat $t = 2.0s$!
- Tentukan arah arus yang mengalir melalui resistor!

Jawab:

emf Induksi Saat $t = 2$

$$\varepsilon_{ind} = -\frac{d\Phi_B}{dt}$$

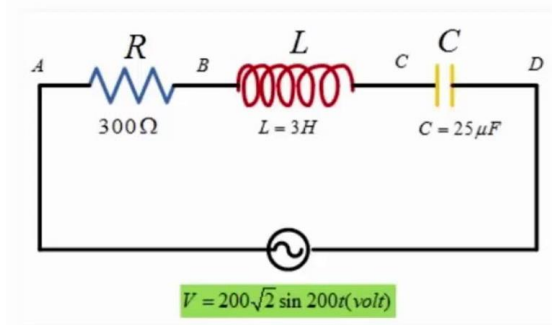
$$\varepsilon_{ind} = -\frac{d(13t^2 + 4t + 8)}{dt}$$

$$\varepsilon_{ind} = -(26t + 4)$$

$$\varepsilon_{ind} = -(26 \times 2 + 4)$$

$$\varepsilon_{ind} = -56 \text{ mV}$$

4. Resonance in RLC Circuit, Including Its Phase Analysis



Tentukanlah :

- Besar Reaktansi induktif
- Besar reaktansi kapasitif
- Besar impedansi
- Besar tegangan efektif
- Besar arus efektif
- Besar tegangan ujung-ujung resistor
- Besar tegangan ujung-ujung induktor
- Besar tegangan ujung-ujung kapasitor
- Besar frekuensi resonansi
- Besar daya efektif
- Besar faktor daya
- Beda potensial antara titik A dan C
- Beda potensial antara titik B dan D

n. beda fase

Berdasarkan $V = 200\sqrt{2} \sin(200t)$, diperoleh:

$$\varepsilon_{max} = 200\sqrt{2}$$

$$\omega = 200 \frac{\text{rad}}{\text{s}}$$

Reaktansi Induktif

$$X_L = \omega L$$

$$X_L = 200 \times 3$$

$$X_L = 600 \, \Omega$$

Reaktansi Kapasitif

$$X_C = \frac{1}{\omega C}$$

$$X_C = \frac{1}{200 \times 25 \times 10^{-6}}$$

$$X_C = \frac{1}{5 \times 10^{-3}}$$

$$X_C = 200 \, \Omega$$

Impedansi

$$Z = \sqrt{(X_L - X_C)^2 + R^2}$$

$$Z = \sqrt{(600 - 200)^2 + 300^2}$$

$$Z = \sqrt{400^2 + 300^2}$$

$$Z = 500 \, \Omega$$

Tegangan Efektif

$$V_{rms} = \frac{V_{max}}{\sqrt{2}}$$

$$V_{rms} = \frac{200\sqrt{2}}{\sqrt{2}}$$

$$V_{rms} = 200 \text{ V}$$

Arus Efektif

$$I_{rms} = \frac{V_{rms}}{Z}$$

$$I_{rms} = \frac{200}{500}$$

$$I_{rms} = 0.4 \text{ A}$$

Tegangan Ujung-Ujung Resistor

$$V_{ab} = IR$$

$$V_{ab} = 0.4 \times 300$$

$$V_{ab} = 120 \text{ V}$$

Tegangan Ujung-Ujung Induktor

$$V_{bc} = I X_L$$

$$V_{bc} = 0.4 \times 600$$

$$V_{bc} = 240 \text{ V}$$

Tegangan Ujung-Ujung Kapasitor

$$V_{cd} = I X_c$$

$$V_{cd} = 0.4 \times 200$$

$$V_{cd} = 80 \text{ V}$$

Frekuensi Resonansi

$$f = \frac{1}{2\pi} \sqrt{\frac{1}{LC}}$$

$$f = \frac{1}{2\pi} \sqrt{\frac{1}{3 \times 25 \times 10^{-6}}}$$

$$f \approx 18.3776 \text{ Hz}$$

Faktor Daya

$$\cos \emptyset = \frac{R}{Z}$$

$$\cos \emptyset = \frac{300}{500}$$

$$\cos \emptyset = 0.6$$

Daya Efektif

$$P_{rms} = V_{rms} I_{rms} \cos \emptyset$$

$$P_{rms} = 200 \times 0.4 \times 0.6$$

$$P_{rms} = 48 \text{ Watt}$$

Beda Potensial Antara Titik A dan Titik C

$$V_{ac} = \sqrt{V_{ab}^2 + V_{bc}^2}$$

$$V_{ac} = \sqrt{120^2 + 240^2}$$

$$V_{ac} = \sqrt{72000}$$

$$V_{ac} = \sqrt{2 \times 36 \times 100 \times 10}$$

$$V_{ac} = 120\sqrt{5} \text{ V}$$

Beda Potensial Antara Titik B dan Titik D

$$V_{bd} = V_{bc} - V_{cd}$$

$$V_{bd} = 240 - 80$$

$$V_{bd} = 160 \text{ V}$$

Beda Fase

$$\tan \theta = \frac{X_L - X_C}{R}$$

$$\tan \theta = \frac{600 - 200}{300}$$

$$\tan \theta = \frac{400}{300}$$

$$\theta = 53^\circ$$